UNIT – III

COMPUTE, STORAGE AND IDENTITY SERVICES

1. Introduction:

Cloud computing provides on-demand access to computing resources such as servers, storage, networking, and security over the internet. It focuses on three core pillars of cloud platforms: Compute Services, Storage Services, and Identity & Security Services, which together enable scalable, secure, and reliable cloud applications.

Cloud Compute Services:

Cloud Compute Services refer to the resources that provide processing power to run applications and workloads in the cloud. These services allow users to create and manage virtual machines, bare metal servers, and scalable compute environments without owning physical hardware.

Key features:

1. Autoscaling

2. High availability

3 Instance grouping

4. Enabling applications to dynamically adjust resources based on demand.

Cloud Storage Services:

Cloud Storage Services provide data storage solutions that allow users to store, retrieve, and manage data over the cloud. These services eliminate the need for local storage infrastructure and ensure high durability and availability of data. Cloud storage is offered in different forms, such as block storage, object storage, and file storage, along with backup and snapshot mechanisms for data protection and disaster recovery.

Identity and Security Services:

Identity and Security Services manage authentication, authorization, and protection of cloud resources. Identity and Access Management (IAM) ensures that only authorized users and services can access cloud resources by using users, groups, roles, and policies. Cloud security services further protect data using encryption, key management, and secrets management to prevent unauthorized access and data breaches.

Importance of These Services in Cloud Platforms:

Compute, storage, and identity services form the **foundation of cloud platforms**. They are essential for the effective development, deployment, and management of cloud-based applications. These services are important because they:

- **Enable scalability and flexibility** by allowing applications to scale resources up or down based on workload demand.
- **Provide cost-effective resource utilization** through pay-as-you-use pricing, eliminating the need for large upfront infrastructure investments.
- **Ensure data security and access control** by managing user identities, permissions, and protecting data using security mechanisms.
- **Support high availability and fault tolerance** by distributing workloads and data across multiple servers and locations.
- **Simplify infrastructure management** by automating resource provisioning, monitoring, and maintenance.

Without compute, storage, and identity services, cloud platforms would not be capable of supporting modern, large-scale, secure, and highly available applications.

Real-World Examples

- Compute Services are used to host web applications, mobile app backends, and enterprise servers, where processing power is required to handle user requests and business logic.
- Storage Services are used to store user data, images, videos, documents, databases, and backup files, ensuring high durability and easy access from anywhere.
- Identity & Security Services are used to control access to enterprise applications and cloud resources by managing user authentication, authorization, secure login, and permission policies.

2. Compute Services:

Compute services provide the processing power required to run applications, services, and workloads in a cloud environment. These services allow users to deploy and manage applications without owning or maintaining physical hardware. Cloud compute services support key features such as scalability, high availability, fault tolerance, and cost efficiency, making them a core component of modern cloud platforms.

Compute services are implemented using various models to meet different performance and management requirements. The major components of compute services include Virtual Machines, Bare Metal Servers, Autoscaling, and Instance Groups, each serving a specific purpose in cloud infrastructure.

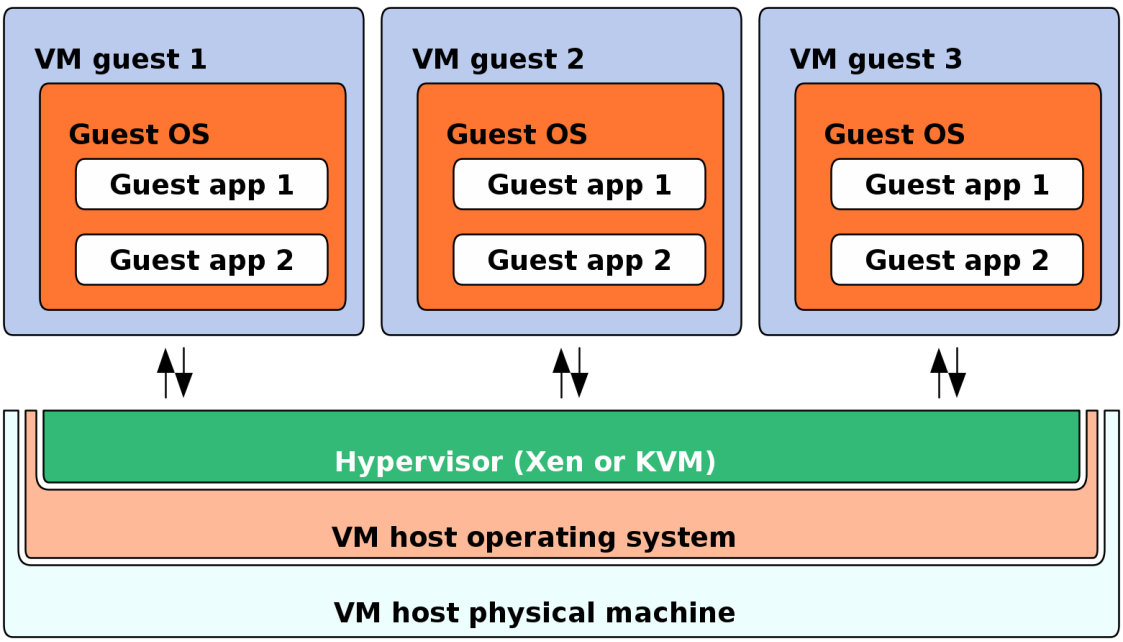
- **Virtual Machines (VMs)** provide flexible, virtualized computing environments that allow multiple operating systems and applications to run on shared physical hardware.
- **Bare Metal Servers** offer direct access to physical hardware, delivering high performance for compute-intensive and latency-sensitive workloads.

- **Autoscaling** enables automatic adjustment of compute resources based on workload demand, ensuring optimal performance while minimizing cost.
- **Instance Groups** manage collections of virtual machines together to improve scalability, reliability, and high availability of applications.

Together, these compute service components form the backbone of cloud application deployment and execution.

2.1 Virtual Machines:

A Virtual Machine (VM) is a software-based emulation of a physical computer that runs an operating system and applications by sharing physical hardware resources through virtualization technology.



<u>Architecture of Virtual Machines:</u>

A **virtual machine architecture** defines how physical hardware resources are shared and managed to run multiple virtual machines efficiently. It consists of the following components:

Host Operating System (Host OS):
The operating system installed on the physical machine that directly interacts with and manages the underlying hardware resources such as CPU, memory, storage, and network.

Hypervisor:
A virtualization layer that creates, manages, and allocates physical hardware resources to multiple virtual machines while ensuring isolation between them.

Guest Operating System (Guest OS):
The operating system installed and running inside a virtual machine, which operates as if it were running on a dedicated physical computer.

Applications:
Software programs that run on the guest operating system to perform specific user or business tasks.

Types of Hypervisors:

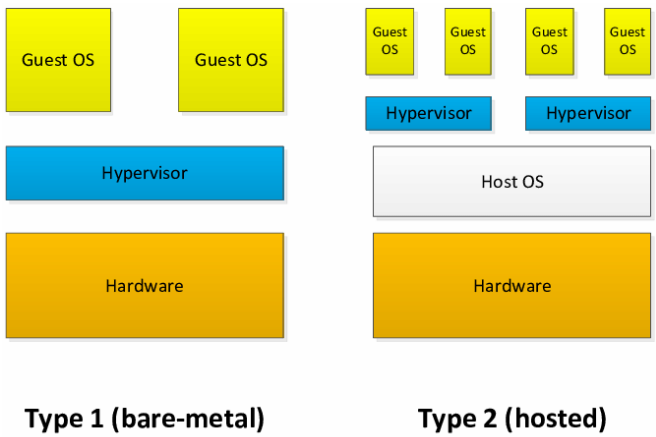
A hypervisor is software that enables virtualization by allowing multiple virtual machines to run on a single physical system. Based on how they interact with hardware, hypervisors are classified into Type 1 and Type 2.

Type 1 – Bare Metal Hypervisor

- Runs directly on physical hardware without a host operating system
- Provides high performance and strong security
- Reduces overhead and improves resource efficiency
- Commonly used in cloud data centers and enterprise environments

Type 2 – Hosted Hypervisor

- Runs on top of a host operating system
- Easy to install and use
- Suitable for testing, learning, and development purposes
- Introduces more overhead compared to Type 1 hypervisors



Advantages of Virtual Machines

- Efficient utilization of hardware resources
- Strong isolation between applications
- Easy scalability and flexibility
- Cost-effective compared to physical servers
- Supports multiple operating systems on the same hardware

Limitations of Virtual Machines

- Performance overhead due to virtualization layer
- Requires careful resource management
- Slower performance compared to bare metal servers

Use Cases

- Hosting web and application servers
- Software development and testing environments
- Running enterprise applications
- Deploying cloud-based services

2.2 Bare Metal Servers

A Bare Metal Server is a dedicated physical server allocated to a single user without virtualization, providing direct access to hardware resources such as CPU, memory, storage, and network interfaces.

When to Use Bare Metal Servers

Bare metal servers are preferred:

- When high performance is critical
- For sensitive or regulated data processing
- For applications with strict latency requirements
- When direct hardware control is required
- When virtualization is not suitable

Advantages of Bare Metal Servers

- Delivers maximum performance
- No performance loss due to virtualization overhead
- Provides strong security and physical isolation
- Offers predictable and consistent performance
- Ideal for resource-intensive and latency-sensitive workloads

Use Cases

- High-Performance Computing (HPC)
- Big Data Analytics and Machine Learning
- Online gaming servers

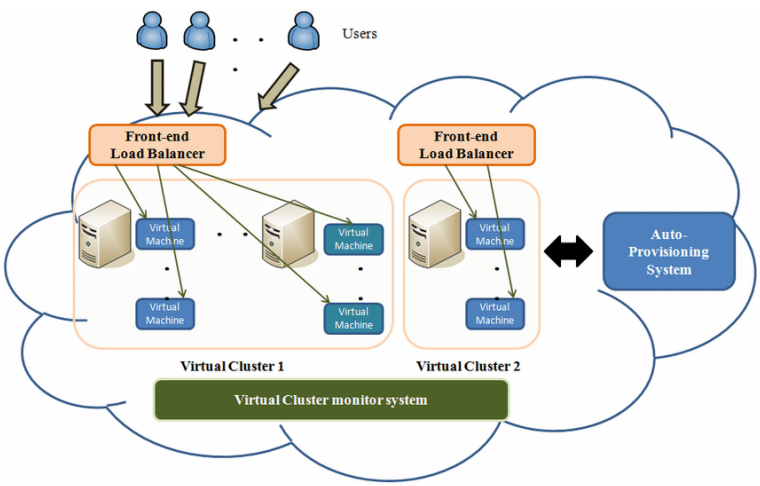
- Financial transaction processing systems
- Scientific simulations and research workloads

Difference between Bare Metal Server and Virtual Machine

Aspect	Bare Metal Server	Virtual Machine
Hardware Access	Direct access to physical hardware	Access through virtualization
Performance	Very high	Moderate
Virtualization Overhead	None	Present
Isolation	Physical isolation	Logical isolation
Scalability	Limited and manual	High and automatic
Cost	Higher	Cost-effective
Provisioning Time	Slower	Faster
Resource Flexibility	Fixed hardware configuration	Flexible and dynamic
Maintenance	User-managed	Cloud provider-managed
Fault Tolerance	Limited	High with autoscaling & instance groups
Use Environment	High-performance workloads	General-purpose workloads

2.3 Autoscaling

Autoscaling is a cloud feature that automatically increases or decreases compute resources such as virtual machines based on real-time workload demand and predefined scaling policies.

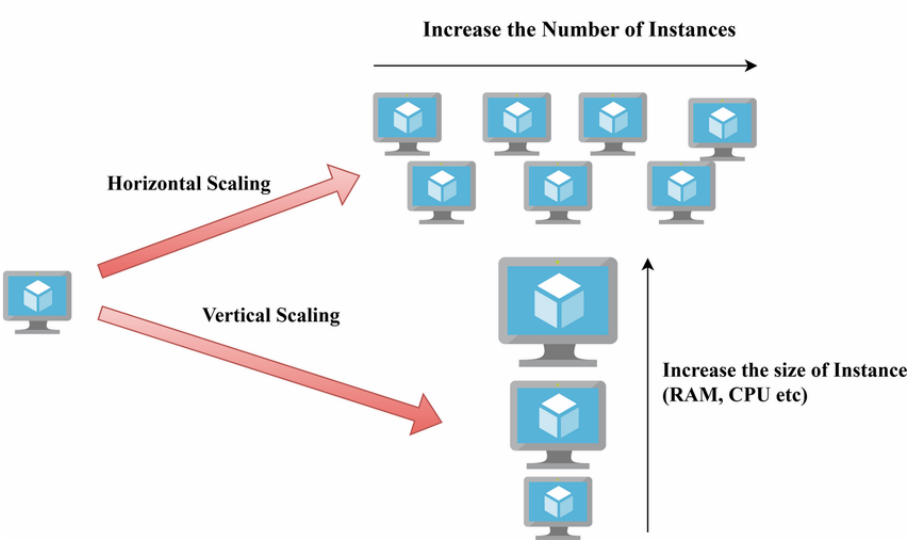


Need for Autoscaling

Autoscaling is required in cloud environments to:

- Handle traffic fluctuations during peak and off-peak hours
- Reduce operational cost by avoiding over-provisioning of resources
- Maintain consistent application performance
- Avoid manual intervention in resource management
- Ensure business continuity during sudden traffic spikes
- Improve user experience by reducing response time.

Types of Scaling



Horizontal Scaling, also known as *scale out* or *scale in*, involves adding or removing virtual machine instances based on workload demand. This type of scaling improves fault tolerance and availability by distributing the workload across multiple instances. Horizontal scaling is widely used in web applications and cloud-native systems where traffic can vary significantly.

Vertical Scaling, also known as *scale up* or *scale down*, involves increasing or decreasing the resources such as CPU, RAM, or storage of an existing virtual machine. It is simpler to implement compared to horizontal scaling, as it does not require managing multiple instances. However, vertical scaling is limited by the physical hardware capacity and may require downtime during resource upgrades.

Benefits of Autoscaling

- Ensures high availability of applications
- Provides cost optimisation through dynamic resource usage
- Improves performance and responsiveness

- Enables automatic resource management

2.4 Instance Groups

An Instance Group is a collection of virtual machine instances that are managed together as a single logical unit to simplify deployment, scaling, and availability management in a cloud environment.

Types of Instance Groups

Managed Instance Groups (MIGs)

- Automatically managed by the cloud platform
- Support autoscaling based on workload demand
- Provide auto-healing by replacing failed instances
- Use a uniform VM configuration (same image and settings)
- Commonly used in production environments

Unmanaged Instance Groups

- Instances are managed manually by the user
- Do not support autoscaling or auto-healing
- Allow flexible and heterogeneous VM configurations
- Mainly used for legacy or custom setups

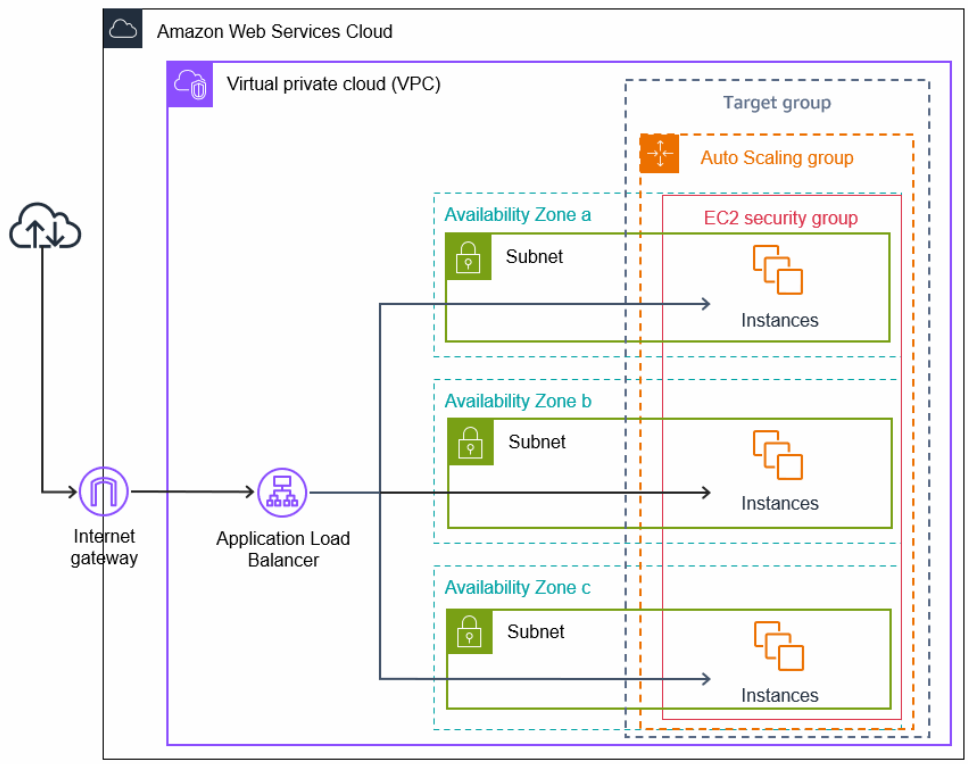
Role in High Availability

Instance groups play a key role in ensuring high availability by:

- Distributing VM instances across multiple zones or regions
- Automatically detecting and replacing failed instances
- Maintaining continuous application availability even during failures

Relation with Autoscaling and Load Balancer

- Autoscaling dynamically increases or decreases the number of VM instances within an instance group based on performance metrics
- Load balancers distribute incoming traffic evenly across instances in the group
- Instance groups act as backend resources that connect autoscaling and load balancing mechanisms



Advantages of Instance Groups

- Simplified virtual machine management
- Improved application reliability
- Easy and automatic scaling
- Enhanced fault tolerance
- Better integration with autoscaling and load balancing

3. Cloud Storage Services:

Cloud Storage is a service that allows users to store, manage, and access data remotely via cloud infrastructure, rather than relying on local or on-premises storage devices.

Characteristics of Cloud Storage

Scalability: Cloud storage allows storage capacity to be increased or decreased dynamically based on application requirements, without service interruption.

Durability: Data is replicated across multiple servers and locations, ensuring protection against hardware failures and data loss.

Availability: Stored data can be accessed anytime and from anywhere through the internet, ensuring continuous access to information.

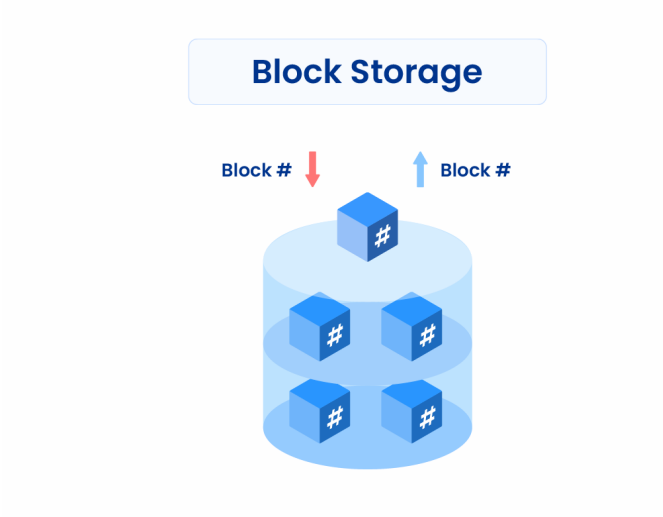
Cost Efficiency: Cloud storage follows a pay-as-you-use pricing model, which reduces capital investment and operational costs.

3.1 Block Storage

Block Storage is a cloud storage method in which data is stored in fixed-size blocks, each with a unique identifier and functioning like a raw hard disk that can be attached to a virtual machine.

How Block Storage Works

In block storage, data is first divided into equal-sized blocks. Each block is stored independently and assigned a unique address. These blocks are then attached to virtual machines as storage volumes, similar to physical disks. The operating system manages the file system, controls data placement, and handles read and write operations.



Advantages of Block Storage

- Provides high performance with low latency
- Ideal for transaction-heavy workloads
- Supports random read and write operations
- Offers reliable and consistent performance
- Suitable for structured data storage

Limitations of Block Storage

- Not suitable for unstructured data such as images and videos
- Scalability is limited compared to object storage
- Requires file system management by the operating system

Use Cases

- Databases requiring fast read/write access
- Operating system disks for virtual machines

- Enterprise applications with transactional workloads

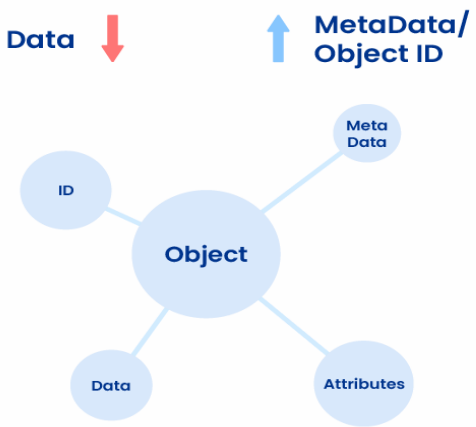
3.2 Object Storage

Object Storage is a cloud storage model that stores data as objects, where each object contains the actual data, metadata, and a unique identifier. Unlike traditional storage systems, object storage does not use a hierarchical file structure.

Structure of Object Storage

An object storage system is organized using the following components:

- **Bucket:**
  A logical container used to store and organize objects.
- **Object:**
  The actual data or file, such as an image, video, or document.
- **Metadata:**
  Descriptive information about the object, such as size, creation date, content type, and access permissions.



Advantages of Object Storage

- Highly scalable, capable of storing massive amounts of data
- Cost-effective for storing large datasets
- Well-suited for unstructured data
- Easy data retrieval using unique identifiers

Limitations of Object Storage

- Higher latency compared to block storage
- Not suitable for transactional or database workloads

- Limited support for random read/write operations

Use Cases

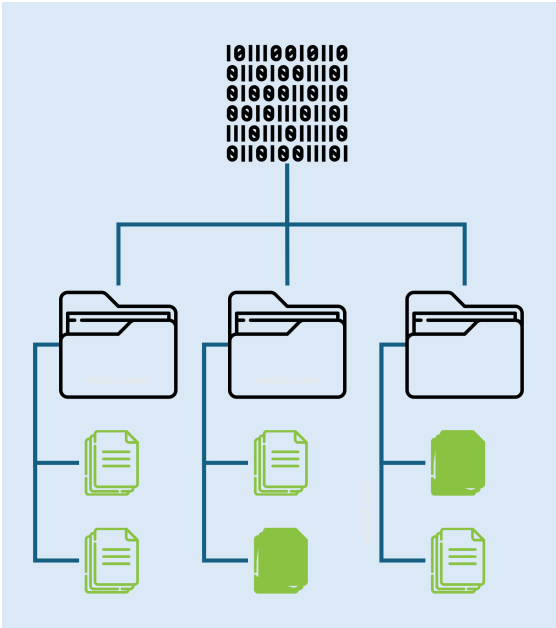
- Storage of images and videos
- Media streaming files
- Backups and archival data
- Static website content

3.3 File Storage

File Storage is a cloud storage method that stores data in a hierarchical structure using files and folders, similar to traditional file systems used in operating systems.

Working Principle

In file storage, data is stored as individual files that are organized into directories and subdirectories within a hierarchical structure. Each file is uniquely identified using a file path, which helps in locating and accessing the data. Cloud-based file storage enables multiple users or systems to access and share files simultaneously over a network. To maintain data integrity and security, access permissions, authentication mechanisms, and file locking techniques are used to control access and prevent conflicts during simultaneous file operations.



Advantages of File Storage

- Simple and familiar file-folder structure
- Easy to manage and organize data
- Supports shared access by multiple users

- Compatible with many applications and operating systems
- Suitable for structured and semi-structured data

Limitations of File Storage

- Scalability is limited compared to object storage
- Performance may reduce with very large datasets
- Not ideal for highly distributed cloud-native applications

Use Cases

- Shared file systems in organizations
- Content management systems
- Collaboration platforms and document sharing
- Home directories and enterprise file sharing
- Web application file storage

Feature	**Block Storage**	**Object Storage**	**File Storage**
Data Structure	Blocks	Objects	Files & Folders
Access Method	Attached as disk	Accessed via APIs	Accessed via file system
Performance	High	Moderate	Moderate
Scalability	Limited	Very High	Moderate
Data Type	Structured	Unstructured	Semi-structured
Latency	Low	Higher	Medium
Metadata Support	Limited	Extensive	Basic
File Hierarchy	No	No	Yes
Consistency	Strong	Eventual	Strong
Cost	Higher	Low	Medium
Typical Use	Databases, OS disks	Media, backups, archives	Shared file access

3.4 Backup & Snapshot Management

Backup and snapshot management are essential cloud storage mechanisms used to protect data, ensure recoverability, and support disaster recovery strategies.

A Backup is a complete copy of data stored in a separate storage location. It is used to restore data in case of system failure, data corruption, or accidental deletion. A Snapshot is a point-in-time copy of data that captures the current state of a disk, volume, or system. Snapshots are usually incremental and depend on the original data source.

Difference Between Backup and Snapshot

Aspect	**Backup**	**Snapshot**
Storage Location	Separate storage	Same storage system
Creation Speed	Slower	Faster
Data Size	Full copy	Incremental
Impact on System	Minimal after completion	Very low
Retention Period	Long-term	Short-term
Recovery Type	Long-term recovery	Quick recovery

Importance in Disaster Recovery

- Protects data from accidental deletion and corruption
- Enables fast system restoration during failures
- Ensures business continuity and service availability

4. Identity and Access Management (IAM):

Identity and Access Management (IAM) is a framework of policies, technologies, and services used to identify users and systems, authenticate them, and control their access to cloud resources in a secure and managed manner.

Need for IAM in Cloud

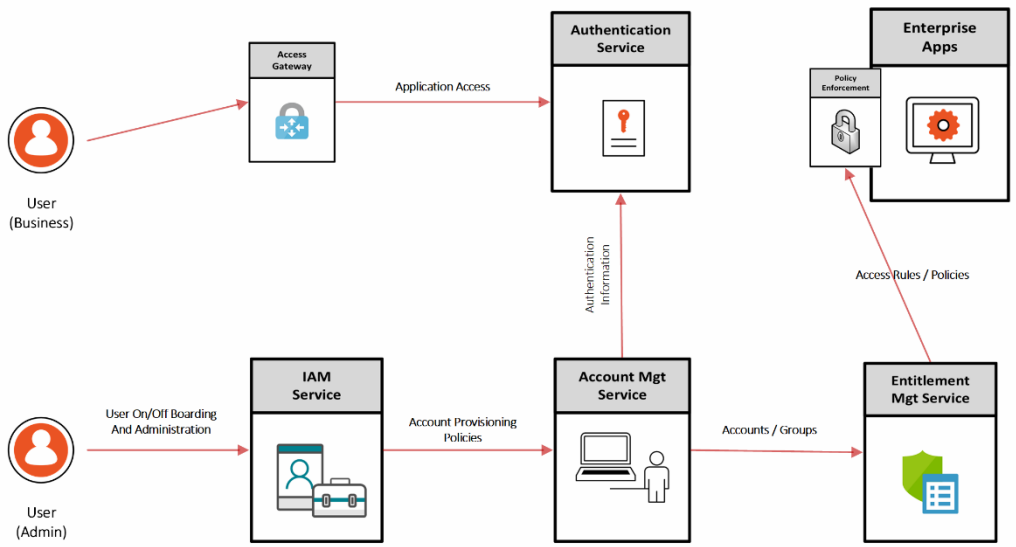
IAM is required in cloud platforms to:

- Prevent unauthorized access to cloud resources

- Enforce authentication and authorization mechanisms
- Implement Role-Based Access Control (RBAC) for fine-grained permissions
- Ensure data security, privacy, and regulatory compliance
- Manage users, applications, and services at large scale
- Reduce security risks, insider threats, and accidental misuse
- Support secure access across multiple services and environments

4.2 IAM Components

IAM consists of several core components that work together to ensure secure, controlled, and auditable access to cloud resources. The major IAM components are Users, Groups, Roles, and Policies.



<u>Users</u>

A User is an individual identity that represents a person or application requiring access to cloud resources. Each user is uniquely identified and authenticated before access is granted.

Key Characteristics:

- Has a unique identity (username or ID)
- Can be assigned permissions directly or through groups
- Uses credentials such as passwords, access keys, or MFA

Example:
A system administrator, developer, or DevOps engineer accessing cloud services using a username and password or access keys.

Use in Cloud: Users are mainly used for human access to cloud management consoles and services.

Groups
A Group is a logical collection of users created to simplify permission management in cloud environments.

Benefits of Grouping Users:

- Simplifies access control by managing permissions centrally
- Reduces administrative effort when users join or leave
- Ensures consistent permission assignment
- Improves overall security management
- Minimizes configuration errors

Example:
A developer group with permissions to access compute and storage services, where all developers inherit the same access rights.

Importance: Groups help enforce role-based access control (RBAC) efficiently at scale.

Roles
A Role is an identity that contains a set of permissions and can be assumed temporarily by users, applications, or cloud services.

Key Characteristics:

- Does not have permanent credentials
- Uses temporary security credentials
- Can be assumed dynamically

Role vs User:

Feature	User	Role
Identity Type	Permanent	Temporary
Credentials	Long-term	Temporary
Primary Use	Human access	Application/service access
Credential Expiry	Manual	Automatic
Security Level	Moderate	High

Temporary Credentials:

Roles use temporary security credentials that automatically expire after a short duration. This significantly reduces the risk of credential misuse and improves overall cloud security.

Use in Cloud: Roles are widely used for application-to-application access, cross-service communication, and secure automation.

Policies

A Policy is a structured document that defines permissions by specifying which actions are allowed or denied on specific cloud resources.

Policy Structure:

A policy typically contains the following elements:

- Effect: Specifies whether the action is *Allowed* or *Denied*
- Action: Defines the operations that can be performed
- Resource: Specifies the cloud resources affected
- Condition: Optional rules that control when the policy applies

Types of Policies:

- **Managed Policies:** Predefined or reusable policies managed by the cloud provider or administrators.
- **Inline Policies:** Policies embedded directly within a user, group, or role for specific use cases.
- **Resource-Based Policies:** Policies attached directly to resources to control who can access them.

Importance of IAM Components

- Fine-grained access control
- Centralized security management
- Scalability in large cloud environments
- Compliance with security and audit requirements
- Reduced risk of unauthorized access

4.3 Federation

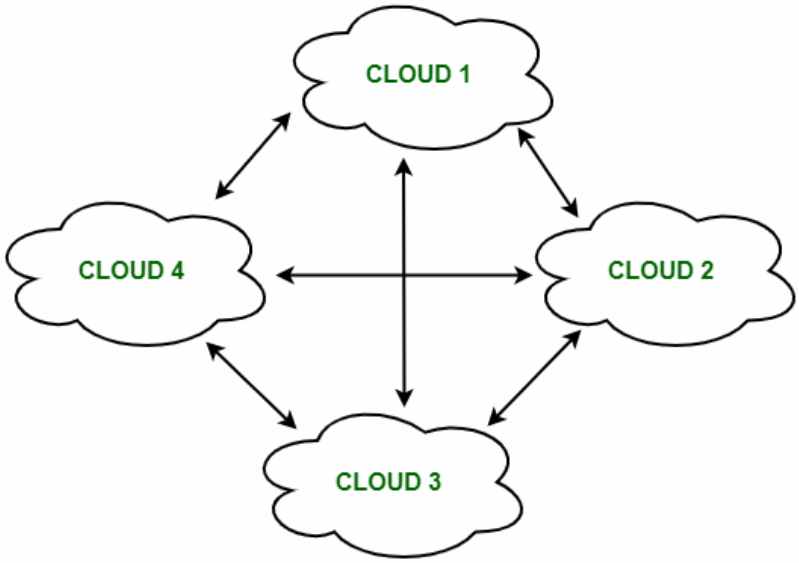
Federation is an identity management mechanism that allows users to access cloud resources using external identities without creating separate user accounts in the cloud platform.

Example: In an enterprise environment, an employee logs into cloud services using corporate credentials through Single Sign-On (SSO), without creating a separate cloud user account.

How Federation Works

Federation operates through a trust-based authentication process:

- The user first authenticates with an external identity provider such as an organization's directory service.
- A trust relationship is established between the cloud service provider and the external identity provider.
- Upon successful authentication, the cloud platform issues temporary security credentials.
- The user then gains controlled access to cloud resources based on assigned permissions.



Benefits of Federation

- Enables Single Sign-On (SSO) across cloud and enterprise applications
- Reduces the need for multiple usernames and passwords
- Improves user experience and productivity
- Enhances security by avoiding long-term credentials

5. Cloud Security

Cloud security refers to the set of technologies, policies, standards, and controls used to protect cloud-based systems, data, and infrastructure from unauthorized access, data breaches, and cyber threats. It ensures the confidentiality, integrity, and availability of data stored and processed in cloud environments.

5.1 Encryption

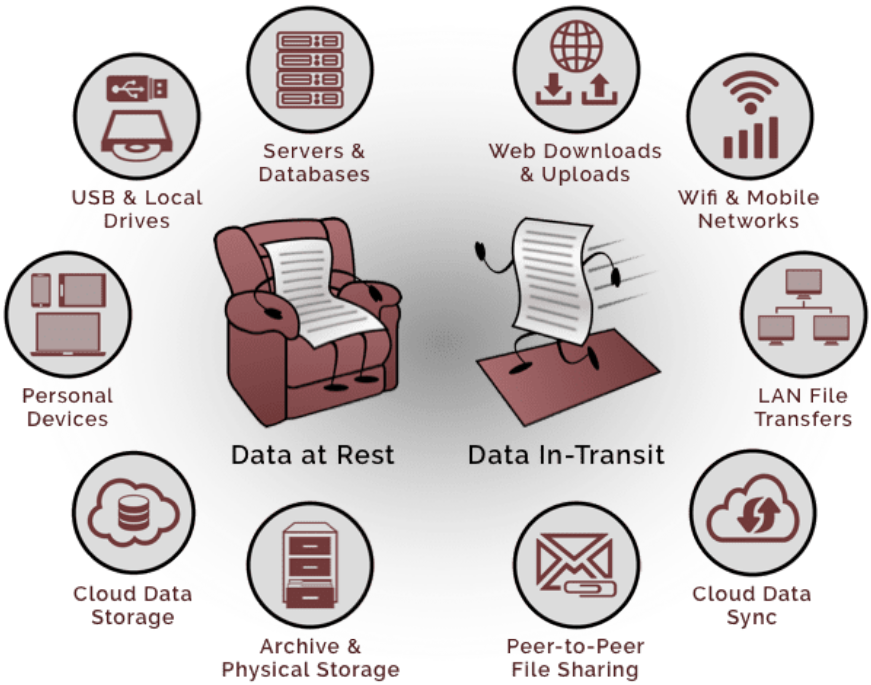
Encryption is the process of converting data into an unreadable or encoded format using cryptographic algorithms, so that only authorized users with the correct decryption keys can access the original data.

Data at Rest

Data at rest refers to data that is stored in cloud environments, such as databases, virtual machine disks, object storage, and backups. Encryption of data at rest ensures that stored data remains secure even if storage devices are compromised or accessed without authorization. It protects sensitive information such as personal data, financial records, and business documents.

Data in Transit

Data in transit refers to data that is being transmitted over networks between users, applications, and cloud services. Encryption during transmission protects data from interception, eavesdropping, and man-in-the-middle attacks, ensuring secure communication between endpoints.



Importance of Encryption

Encryption plays a critical role in cloud security because it:

- Protects sensitive and confidential data
- Prevents unauthorized access to stored and transmitted information
- Ensures data privacy and confidentiality
- Maintains data integrity by preventing tampering
- Helps meet regulatory and compliance requirements
- Reduces the impact of data breaches and cyberattacks
- Builds trust between cloud service providers and users

5.2 Key Management

Key Management is the process of creating, storing, using, rotating, and deleting cryptographic keys that are used for data encryption and decryption in cloud environments. Proper key management ensures that encrypted data remains secure and accessible only to authorized entities.

Key Lifecycle

The key lifecycle defines how cryptographic keys are securely managed from creation to destruction in a cloud environment. Proper handling of each stage is essential to maintain data confidentiality, integrity, and compliance.

1. Key Generation

Key generation is the process of creating cryptographic keys using secure algorithms and random number generators. Strong key generation ensures that keys are unpredictable and resistant to attacks.

Example: When a new cloud database or storage volume is created, the cloud platform automatically generates a unique encryption key.

2. Key Registration

Key registration involves recording key metadata such as key ID, creation date, purpose, and ownership in the key management system. This helps in tracking and managing keys throughout their lifecycle.

Example: A newly generated key is registered with details specifying whether it is used for storage encryption or network communication.

3. Key Storage

After registration, keys are stored securely in a centralized and protected key management system or key vault. Access to stored keys is tightly controlled using policies and permissions.

Example: Encryption keys are stored in a cloud key vault rather than being embedded in application code.

4. Key Distribution & Installation

In this stage, keys are securely distributed and installed to authorized applications, services, or systems that require them. Secure channels are used to prevent interception.

Example: A virtual machine receives an encryption key securely to encrypt its attached storage disk.

5. Key Use

Key use refers to the actual encryption and decryption operations performed using the key. Access to key usage is restricted to authorized entities only.

Example: A storage service uses an encryption key to encrypt data before writing it to disk and decrypts it when accessed by an authorized user.

6. Key Rotation

Key rotation involves periodically replacing old keys with new ones to minimize security risks. Regular rotation limits the impact of a compromised key.

Example: An organization rotates encryption keys every 90 days as part of its security policy.

7. Key Backup & Recovery

Keys are backed up securely to prevent permanent data loss in case of accidental deletion or system failure. Recovery mechanisms ensure business continuity.

Example: A backup copy of an encryption key is stored securely to recover encrypted data during a disaster.

8. Key Revocation & Suspension

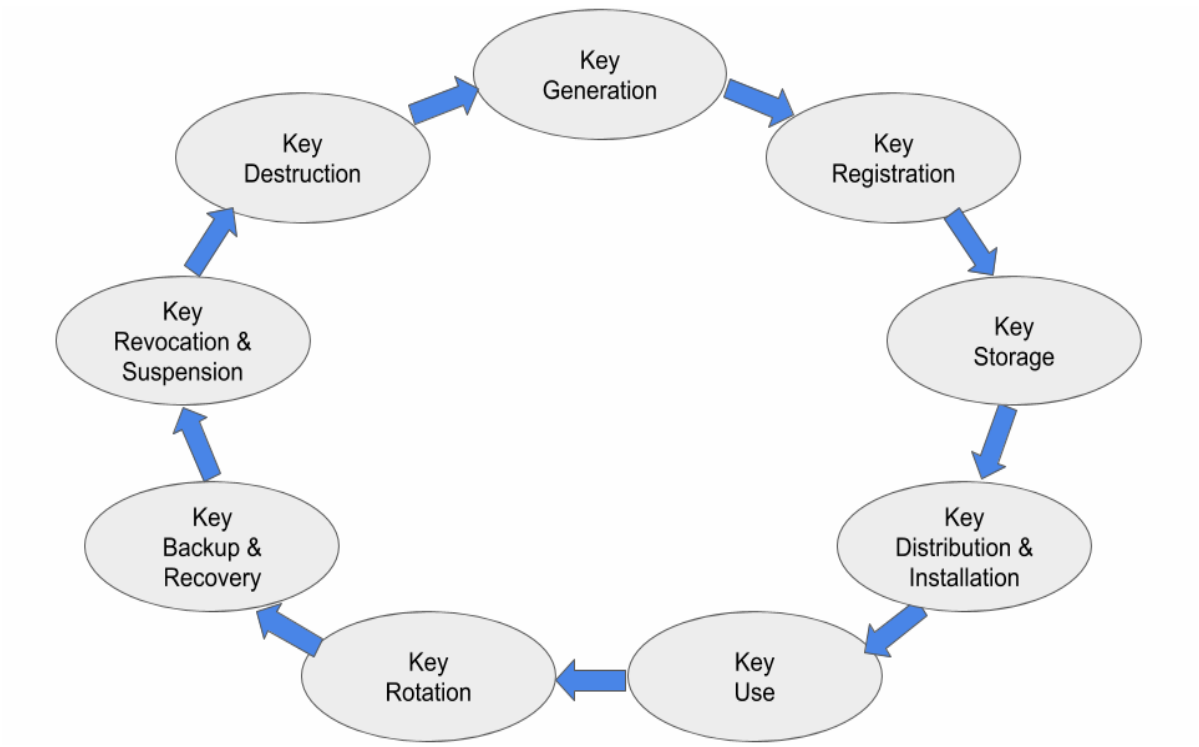
If a key is compromised, expired, or misused, it is revoked or temporarily suspended to prevent further access.

Example: When an employee leaves the organization, the associated encryption keys are revoked immediately.

9. Key Destruction

Key destruction is the final stage, where keys are permanently deleted so they cannot be recovered or reused. This ensures encrypted data cannot be accessed once the key is destroyed.

Example: Encryption keys related to a decommissioned application are permanently destroyed.



5.3 Secrets Management

Secrets Management is the practice of securely storing, accessing, and managing sensitive credentials required by applications, users, and services in a cloud environment. It ensures that confidential information is protected from unauthorized access.

<u>What Are Secrets</u>

- **API Key:** A unique identifier used by applications to authenticate and access cloud services securely.
- **Password:** A confidential string used to verify the identity of a user or system.
- **Database Credentials:** Authentication details required for an application to connect securely to a database.
- **Access Token:** A temporary credential that grants limited authorization to access protected resources.
- **Certificate:** A digital credential used to establish trust and enable secure communication.
- **Private Key:** A secret cryptographic key used to decrypt data or digitally sign requests.

- **Authentication Token:** A generated secret used to validate and maintain a secure user session.

Why Secrets Management Is Required

Secrets management is essential because:

- Hard-coded secrets are insecure and can be easily exposed
- Prevents accidental leakage of credentials through code repositories
- Enables secure access for applications and cloud services
- Supports automatic rotation of secrets without service interruption
- Reduces human involvement in handling sensitive data
- Enhances overall application security

Benefits of Secrets Management

- Improves overall cloud security posture
- Reduces the risk of credential theft and misuse
- Enables fine-grained access control and least-privilege access
- Supports automation and DevOps practices
- Enhances compliance, auditing, and monitoring
- Simplifies secret lifecycle management